Tutorial : 2

① 100 kVA, 1100V/230V, 50 Hz Transformer.

primary: LV side

secondary: HV side

Given data

$r_1 = 0.1\ \Omega$ (HV side winding resistance)

$x_1 = 0.4\ \Omega$ (HV side ~~reactance~~ leakage reactance).

$r_2 = 0.006\ \Omega$ (LV side winding resistance)

$x_2 = 0.01\ \Omega$ (LV side leakage reactance)

a) Referring to HV side.

$$r_2' = r_2 \cdot \frac{N_1^2}{N_2^2} = 0.006 \times \frac{1100^2}{230^2} = 0.1372\ \Omega$$

Eqlnt. winding resistance

$$R_1 = r_1 + r_2' = 0.1\ \Omega + 0.1372\ \Omega = 0.2372\ \Omega$$

$$x_2' = 0.01 \times \frac{1100^2}{230^2} = 0.2287\ \Omega$$

Eqlnt. leakage reactance

$$X_1 = x_1 + x_2' = 0.6287\ \Omega$$

b) Referring to LV side

$$r_1' = r_1 \cdot \left(\frac{N_2^2}{N_1^2}\right)$$

$$= 0.1 \times \frac{230^2}{1100^2} = 4.8719\ m\Omega$$

Eqlnt. winding resistance

$$R_2 = r_1' + r_2 = 0.01047\ \Omega$$

$$x_1' = x_1 \cdot \left(\frac{N_2^2}{N_1^2}\right)$$

$$= 0.0175\ \Omega$$

Eqlnt leakage reactance, $X_2 = x_1' + x_2$

$$= 0.0275\ \Omega$$

2) Given a 230V/6 V transformer.

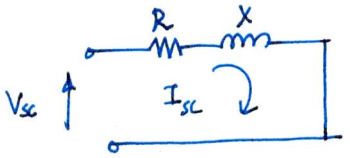
secondary winding is short circuited and voltage applied on primary.

$V_{sc} = 20V$

$I_{sc} = 1A.$

$P_{sc} = 10W$

Equivalent circuit during s.c condition



$R = \frac{P_{sc}}{I_{sc}^2} = \frac{10}{1^2} = 10\Omega$ (referred to primary side).

$Z = \frac{V_{sc}}{I_{sc}} = \frac{20V}{1A} = 20\Omega$

$X = \sqrt{Z^2 - R^2} = 17.32\Omega$ (referred to primary side).

R referred to secondary side $= 10\Omega \times \frac{6^2}{230^2} = 6.8053\,m\Omega$

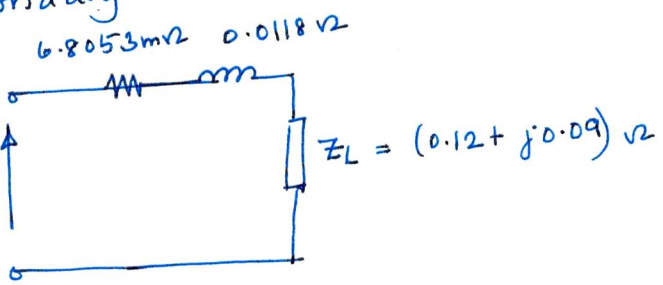
X referred to secondary side $= 17.32 \times \frac{6^2}{230^2} = 0.0118\,\Omega$

A load of $(0.12 + j\,0.09)\,\Omega$ is connected at secondary.

~~Equivalent load impedance referred~~

supply voltage referred to secondary side $= 230V \times \frac{6}{230} = 6V$

Equivalent circuit referred to secondary



$V_{Load} = V_{in} * \frac{Z_L}{Z_L + R + jX}$

$= 6V * \frac{(0.12 + j0.09)}{(6.8053 \times 10^{-3} + j0.0118) + (0.12 + j0.09)}$

$= 5.5346 \angle -1.8878^{\circ}\ V$

3. A transformer unit is given.

50 kVA, 2400V/240V, 60Hz

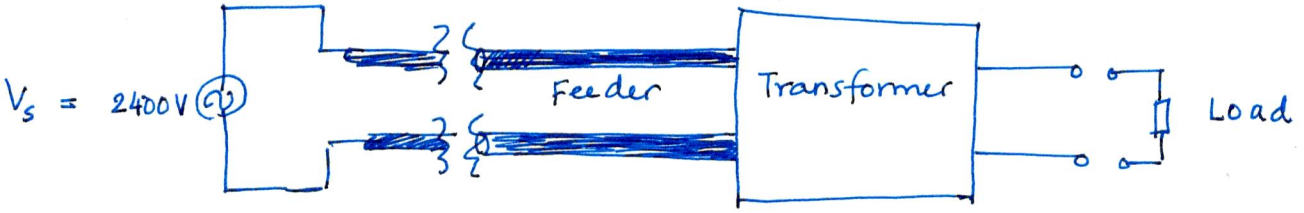
HV side: primary.

LV side: secondary.

$$r_1 + jx_1 = (0.72 + j0.92)\ \Omega$$
$$r_2 + jx_2 = (0.007 + j0.009)\ \Omega$$

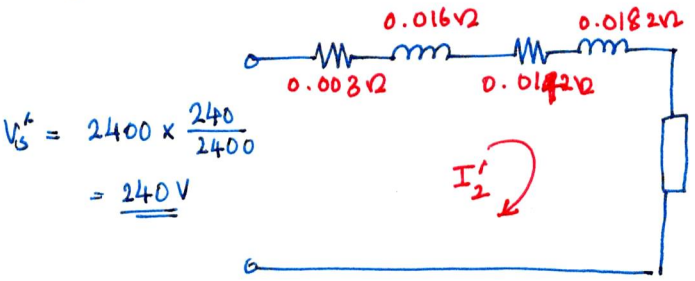
Z_ϕ (referred to LV side) $= (632 + j43.7)\ \Omega$

This transformer unit is used to stepdown voltage at load end of a feeder. The feeder line has an impedance $Z_f = (0.3 + j1.6)\ \Omega$



* Feeder impedance would then come in series with eq. circuit of transformer.

Equivalent circuit referred to LV side.



$$R_{eqlnt} = R = 0.72 \times \frac{240^2}{2400^2} + 0.007$$
$$= 0.0142\ \Omega$$

Equivalent * (referred to LV side)

$$= 0.92 * \frac{240^2}{2400^2} + 0.009$$
$$= 0.0182\ \Omega$$

Z_f (referred to LV side)

$$= (0.3 + j1.6) * \frac{240^2}{2400^2}$$
$$= (0.003 + j0.016)\ \Omega$$

Given that the load draws rated current.

$$\Rightarrow I_2' = \frac{50 \times 10^3}{240} = 208.33$$

$$\overline{I_2'} = 208.33 \angle -36.86^\circ \text{ A}$$

$\cos\phi = 0.8$ lagging.

$$|V_L| = |V_1| - |I_2'| (R\cos\phi + X\sin\phi)$$

where $R = (0.003 + 0.0142) = 0.0172\ \Omega$

$X = (0.016 + 0.0182) = 0.0342\ \Omega$

$$|V_L| = 240 - 208.33\,(0.0172 * 0.8 + 0.0342 * 0.6)$$

$$= 232.8584\ V$$

④ Feeder whose ~~sending~~ impedance $= Z_f = (95 + j360)\ \Omega$

Transformer unit 35000 V / 2400 V

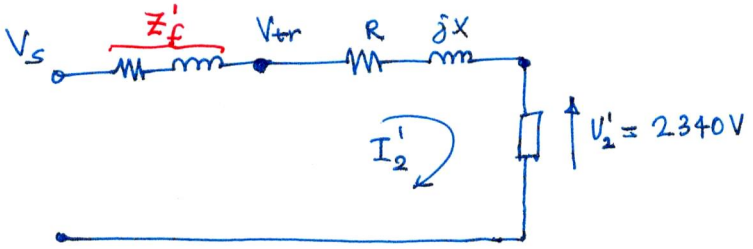
$Z = R + jX = (0.23 + j1.27)\ \Omega$ referred to LV side (secondary side).

Load specifications :- 160 kW, 0.89 pf lead, at 2340 V.

a) Feeder impedance referred to LV side

$$Z_f' = (95 + j360) \times \frac{2400^2}{35000^2} = (0.4467 + j1.6927)\ \Omega.$$

Equivalent circuit referred to LV side



Calculation of load current :-

Given, output power $= P = V_2' I_2' \cos\varphi = 160$ kW

$$\Rightarrow |I_2'| = \frac{160 \times 10^3}{2340 \times 0.89} = 76.827\ A.$$

$\overline{I_2'} = 76.827 \angle 27.1267$ Note that pf is leading. Sign of angle is +ve.

Voltage at HV terminals $= \overline{V_{tr}} = V_2' + I_2' (R + jX)$

$= 2340 \angle 0 + 76.827 \angle 27.1267 * (0.23 + j1.27) = 2313.1856 \angle 2.35$ V

Actual voltage at HV terminals $= 2313.1856 \times \frac{35000}{2400} = 33734$ V ~~33284.5989 V~~

Voltage at sending end of feeder $= \overline{V_s} = \overline{V_{tr}} + \overline{I_2'} (Z_f')$

$$\Rightarrow \overline{V_s} = 2313.1856 \angle 2.35 + 76.827 \angle 27.1267 * (0.4467 + j1.6927)$$

$$= 2293.6732 \angle 5.6606^\circ\ V$$

Actual voltage @ sending end $= 2293 \times \frac{2400}{35000} = 33499.34$ V

Complex power at sending end

$$= \bar{V}_s \, \bar{I}_s^*$$

$$= 2293.6732 \angle 5.6606^\circ \cdot (76.827 \angle 27.1267)^*$$

$$= 2293.6732 \angle 5.6606 \times 76.827 \angle -27.1267$$

$$= (163992.6741 - j\,64486.3738)$$

$\Rightarrow$ Real power = 163992.6741 W

Reactive Power = 64486.3738 VAr

5) 50 kVA, 2200 V / 110 V Transformer.

LV side → secondary

HV side → primary.

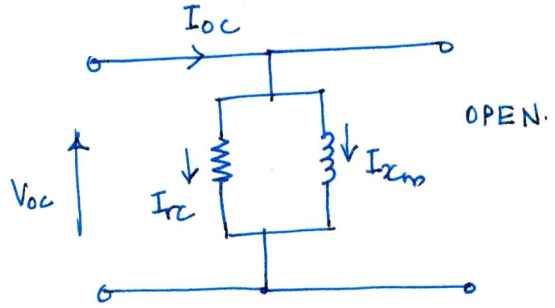
OC Test

Performed on LV side (ie 2° side).

$V_{oc} = 110\ V$

$I_{oc} = 10\ A$

$P_{oc} = 400\ W.$



$\Rightarrow$ core loss resistance $r_c = \frac{V_{oc}^2}{P_{oc}} = \frac{110^2}{400} = 30.25\ \Omega$

Current through ~~magnetizing branch~~ r_c $= \frac{V_{oc}}{r_c} = \frac{110\ V}{30.25} = 3.6364\ A$

Current through mag. branch $I_{xm} = \sqrt{I_{oc}^2 - I_{rc}^2}$

$= \sqrt{10^2 - 3.6364^2} = 9.3154\ A.$

$\Rightarrow x_m = \frac{V_{oc}}{I_{xm}} = 11.8082\ \Omega$

$r_c = 30.25\Omega$ and $x_m = 11.8082\Omega$ → Referred to LV side

$r_c = 30.25 \times \frac{2200^2}{110^2} = 12100\Omega$ and $x_m = 11.8082 \times \frac{2200^2}{110^2} = 4723.28\Omega$

→ Referred to HV side.

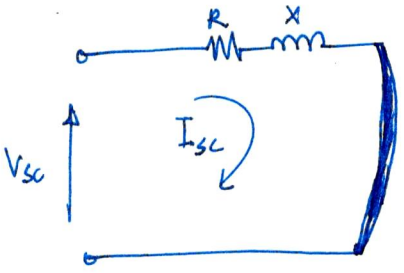
Short Circuit Test

Performed on HV side (ie 1° side).

$V_{sc} = 90\ V$

$I_{sc} = 20.5\ A$

$P_{sc} = 808\ W$



$R = \frac{P_{sc}}{I_{sc}^2} = \frac{808}{20.5^2} = 1.922\ \Omega$ referred to HV side.

$R' = 1.922 \times \left(\frac{110^2}{2200^2}\right) = 0.0048\ \Omega$ referred to LV side.

$$Z_{sc} = \frac{V_{sc}}{I_{sc}}$$

$$= \frac{90}{20.5}$$

$$= 4.3902\ \Omega$$

$$X = \sqrt{Z_{sc}^2 - R^2}$$

$$= 3.9472\ \Omega \quad \text{referred to HV side}$$

$$X' = 3.9472 \times \frac{110^2}{2200^2} = 0.0099\ \Omega \quad \text{referred to LV side.}$$